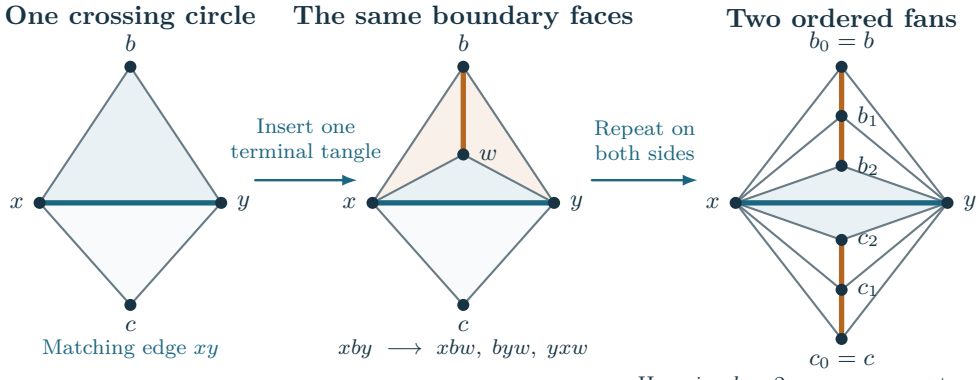

One crossing circle
The same boundary faces
Two ordered fans
b
x
y
c
Matching edge xy
Insert one
terminal tangle
w
xby ⟶ xbw, byw, yxw
Repeat on
both sides
b0 = b
b1
b2
c2
c1
c0 = c
Here j = k = 2; any even counts preserve
the four-endpoint pairing.
retained matching edge
new matching edges
Vertices label the white projection regions; triangles record the crossing-disk
cut. The two shaded triangles beside xy retain the original crossing circle.